

TED PAULO A. FERANIL

MANILA, PH | tedpaulo.feranil@gmail.com | +63 915 7996 213 | [LinkedIn](#) | [GitHub](#)

SUMMARY

An Innovative BS Computer Engineering student with focus on Machine Learning and AI, passionate about developing scalable, real-time AI systems and full-stack solutions. Aims to contribute practical solutions through hands-on experience in AI/ML pipelines, backend development, and data-driven problem solving.

EDUCATION

Polytechnic University of the Philippines – Manila

Manila, PH

Bachelor of Science in Computer Engineering Major in AI/Machine Learning, 2022 – Present (Expected 2026)

Thesis title: EarPrompt: A Real-Time Conversational AI Assistant for Enhancing Verbal Communication

Awards: DOST Scholar (National Scholarship), Consistent President Lister (1st Honor)

EXPERIENCE

NTEK Systems Inc.

Manila, PH

AI Software Developer Intern

July 2024 – September 2024

- Engineered a Retrieval-Augmented Generation (RAG) pipeline using Python, LangChain, Hugging Face Transformers, and FAISS to connect LLMs with private datasets, improving response relevance and reducing query latency.
- Developed scalable Flask-based backend APIs for real-time AI interactions while improving model precision and minimizing hallucinations through iterative testing and prompt engineering.

Department of Science and Technology – PES - ITD

Manila, PH

Programmer/Developer Intern

July 2025 – September 2025

- Contributed to full-stack development of a DOST web initiative using Laravel (PHP), Bootstrap, JavaScript, and MySQL, implementing core backend features including 15+ database migrations, secure authentication, and CRUD operations.
- Debugged and tuned performance to ensure system stability while collaborating with senior developers.

PROJECTS

ConTeX (Handwritten to Latex Code Conversion)

- Designed and developed a Flask-based backend pipeline using Hugging Face Transformers to convert handwritten mathematical expressions into LaTeX.
- Built RESTful APIs for real-time image processing and LaTeX output, increasing recognition accuracy of complex multi-line equations by 25% through optimized image pre-processing.

EarPrompt: Real-Time Conversational AI Assistant

- Led a 4-member team in developing a low-latency conversational AI assistant integrating ElevenLabs, SpeechBrain, and a prompt-engineered LLM to deliver accurate, factually grounded responses.
- Optimized and deployed a Flask-based real-time AI pipeline on Raspberry Pi 5, managing CPU and RAM allocation and synchronizing voice processing with AI inference to maintain sub-2-second latency.

SKILLS & INTERESTS

Programming & Web Development:

- Languages: Python, PHP, JavaScript, HTML, CSS
- Frameworks & Libraries: Flask, Django, Laravel, Bootstrap, Tailwind

AI, Machine Learning & Data Engineering:

- Hugging Face, LangChain, RAG Pipeline, TensorFlow, Scikit-learn, Prompt Engineering

Databases & Data Management:

- MySQL, Firebase (Firestore)

Developer Tools & APIs:

- Git, GitHub, VS Code, Postman, Composer, NPM, REST APIs